Transformations, Sums, and Mixtures: Recap

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Transformations

When $Q(u)$ is hard to find, build X from variables we can already draw.

- **Easy direction**: define $X = g(Y)$, e.g. $\exp(Y)$, $\sum Z_i^2$, $Z/\sqrt{V/k}$.
- **Hard direction**: given $f(x)$, find a g that produces it.
- To prove it works, **show the CDFs match**.

Tips for the hard direction:

- Reduce to a simpler case.
- Write down $f(x)$ or $F(x)$ and look for patterns.
- Condition on an event, then generate the event.
- Change variables.

Example: the Laplace distribution

$$f(x) = \tfrac{1}{2} \exp\{-|x - \theta|\}$$

1. **Shift**: $Y \sim \text{Laplace}(\theta)$ is $X + \theta$ with $X \sim \text{Laplace}(0)$.
2. **Symmetry**: $P(X > 0) = P(X < 0) = 1/2$.
3. **Condition**: $X \mid X > 0$ is $\text{Exp}(1)$, and $X \mid X < 0$ is $-\text{Exp}(1)$.

So $X = SM + \theta$, with a random sign $S = \pm 1$ and $M \sim \text{Exp}(1)$.

In R: `sample(c(-1, 1), n, replace = TRUE) * rexp(n) + theta`.

Sums of random variables

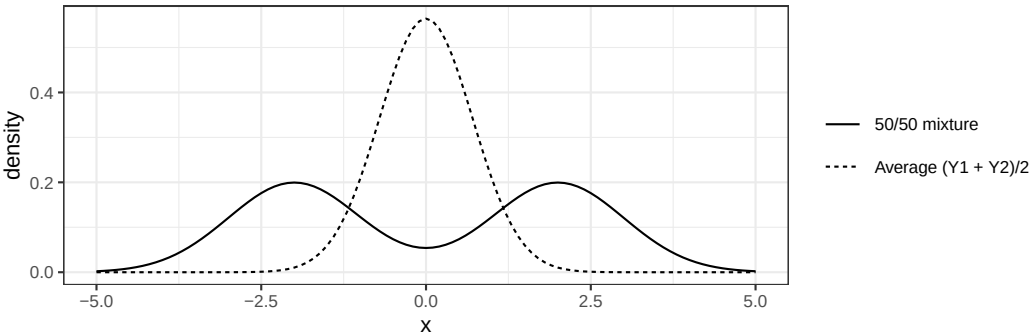
For independent X and Y, $Z = X + Y$ has the **convolution** density

$$f_Z(z) = \int f_X(z - y) f_Y(y)\, dy.$$

- The integral is often hard. **Sampling is easy**: draw X and Y, add them.
- **Gamma**: $\sum_{i=1}^{k} E_i \sim \text{Gamma}(k, \lambda)$ for $E_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, integer k only.
- **Binomial**: $\sum_{i=1}^{m} B_i \sim \text{Bin}(m, p)$ for $B_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.
- Each draw costs k (or m) uniforms, so large sums are slow.

A sum is not a mixture

$Y_1 \sim N(-2, 1)$ and $Y_2 \sim N(2, 1)$, independent.



The **sum** combines the variables. The **mixture** combines the CDFs.

Mixtures

$$F(x) = \sum_{i=1}^{k} a_i F_i(x), \qquad 0 \le a_i \le 1, \quad \sum_{i=1}^{k} a_i = 1$$

- A valid CDF, with density $f(x) = \sum a_i f_i(x)$.
- The quantile function is usually **not** available, even when every Q_i is.

Two readings, two samplers:

- **Latent variable**: draw C with $P(C = i) = a_i$, then draw X from F_C.
- **Random parameter**: draw $\theta \sim G$, then draw X from $F(x; \theta)$.

The random parameter can be continuous: $F(x) = \int F(x; \theta)\, g(\theta)\, d\theta$.